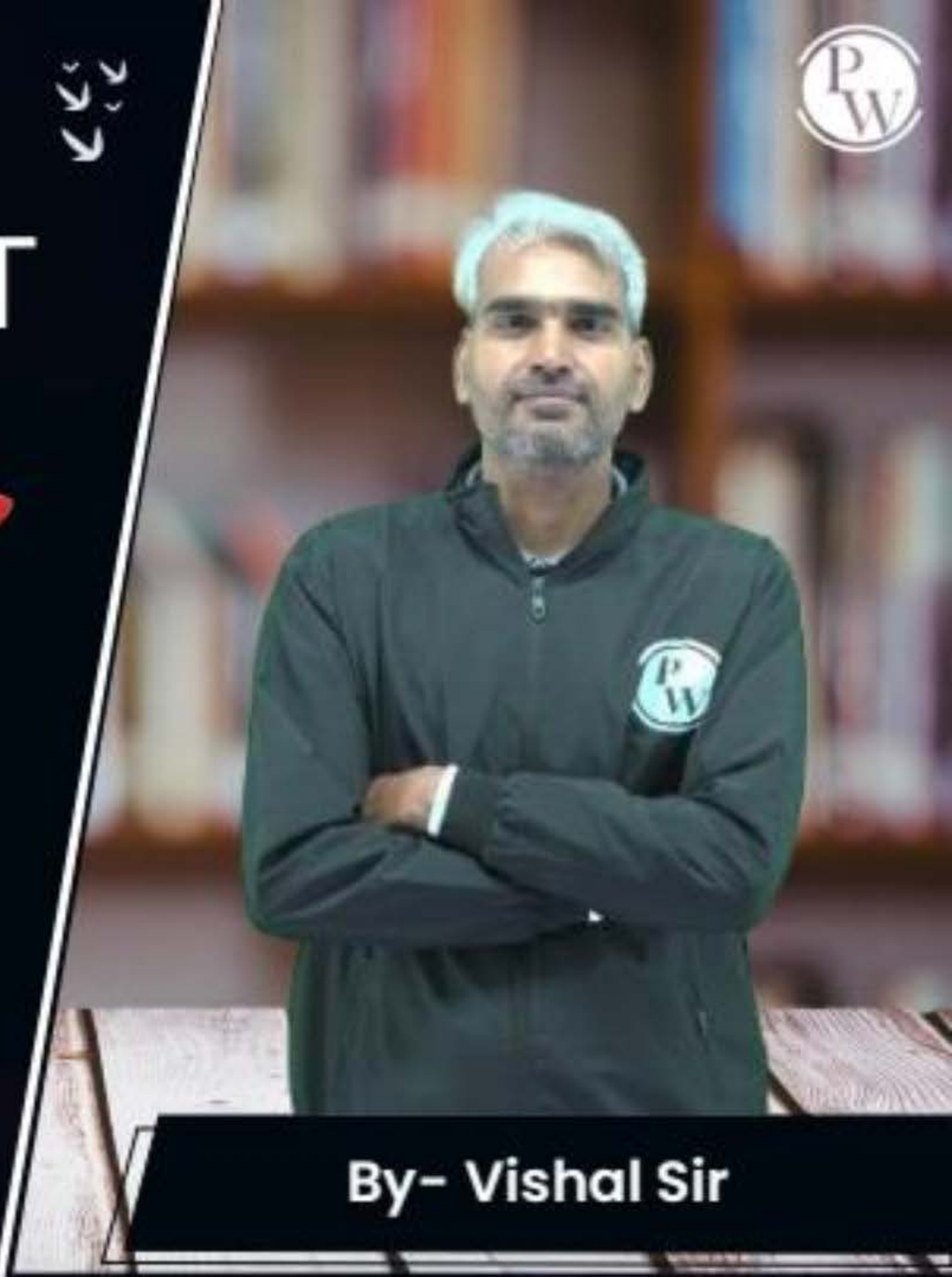
Computer Science & IT



Database Management System

File organization & Indexing

Lecture No. 01



By- Vishal Sir

Recap of Previous Lecture



- Topic: Basic time stamp ordering protocol
- Topic: Thomas write time stamp ordering protocol
- Topic: Wait-die protocol for deadlock avoidance
- Topic: Wound-wait protocol for deadlock avoidance

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Topics to be Covered



- Topic: Database, File and Records
- Topic: Types of Records
- Topic: Blocking Factor
- Topic: Organization of Records
- Topic: Search-key & Type of File
- Topic: Categories of index

Slide



Topic : Database-File-Records

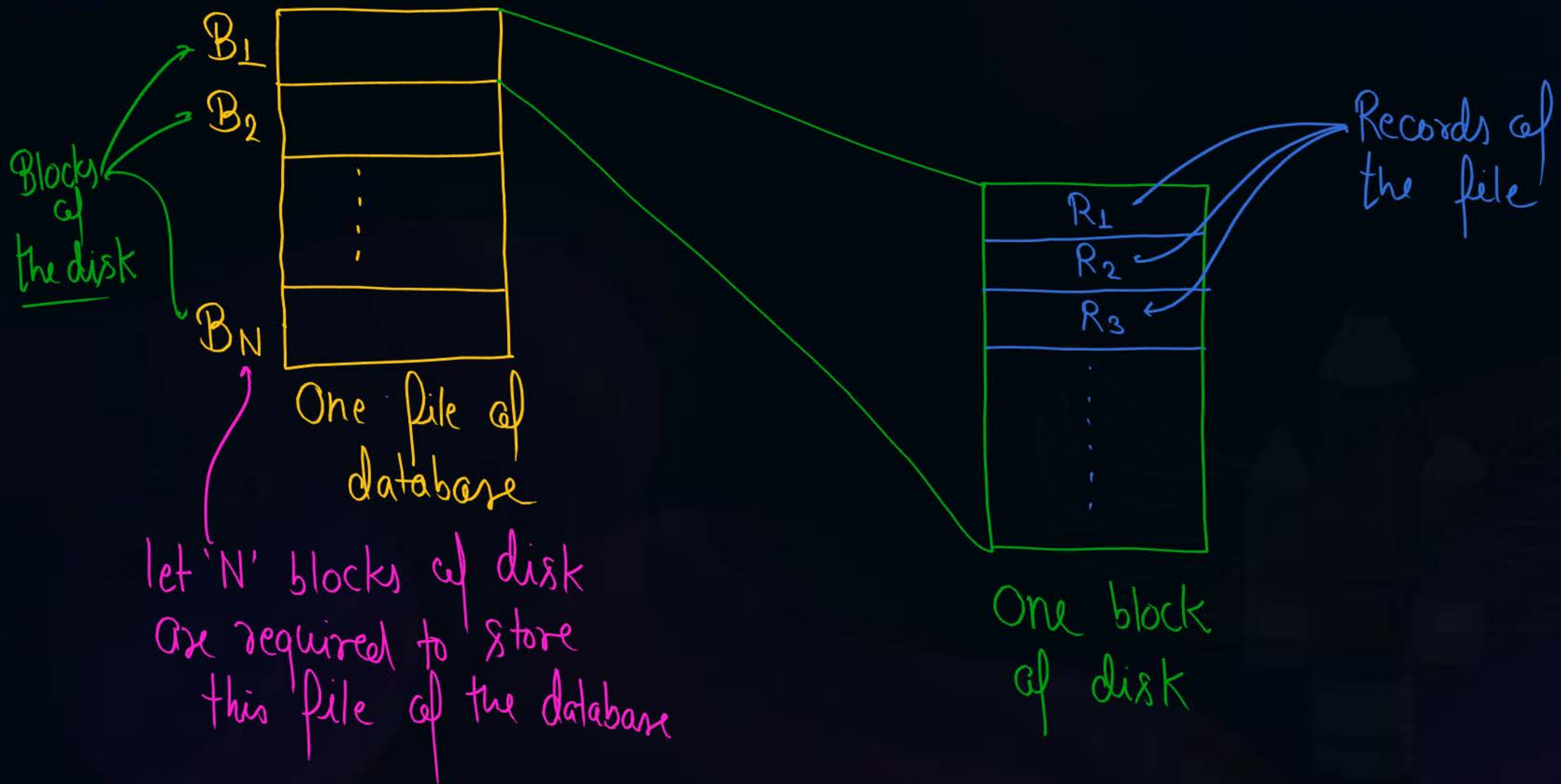
* Database is a Collection of files

files are used to store the records

Records of the file are stored in the blocks of Secondary Memory.



Topic : Database-File-Records





Topic : Types of Records

Two types of records are possible

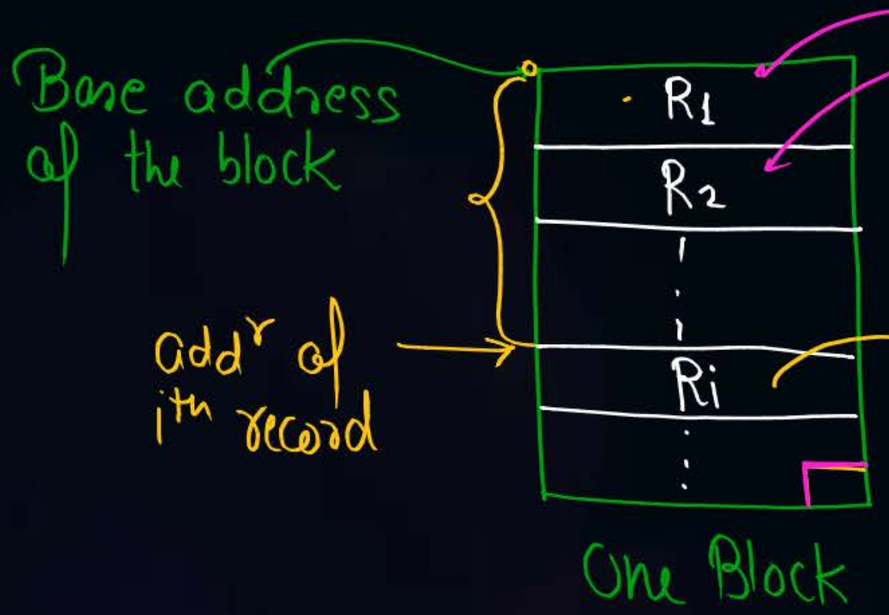
(i) Fixed length records

(ii) Variable length records



Topic : Fixed length records

- When each record of the file is of same size.



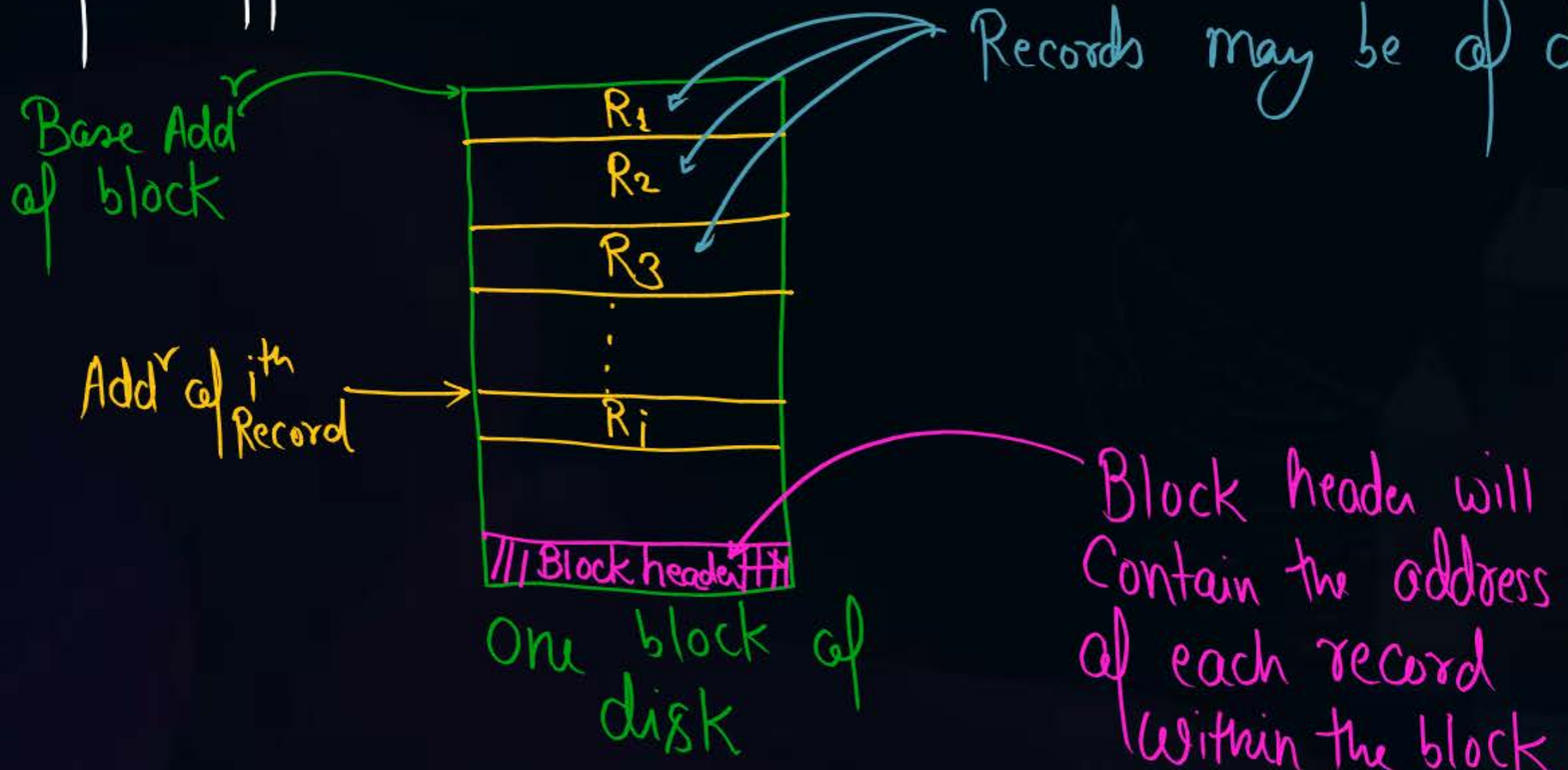
Address of ith Record of the block = Base Addr of block + (i-1) × Record Size

This can be used only if each record of the file is of same size



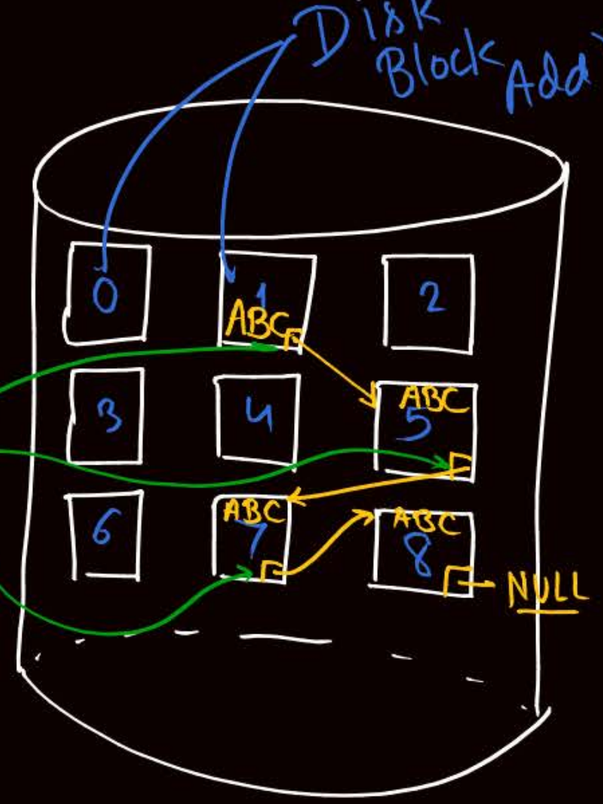
Topic : Variable length records

- In this case records of the file may be of different sizes.



Block header may be required with fixed length records as well in order to provide the address of the next block of the file

eg:



Pointer providing the addr of the next block of the file is called block anchor.

Note:- If nothing is specified about the size of block header in the question, then block head size is considered "Zero".

Note: If block header size is specified in the question, then space available to store the records will be = (Disk block size) − (Block header size)

Effective space available to store the records



Topic : Blocking Factor

* Blocking factor is defined as average number of records per block

$$\text{Blocking (Bf)} = \frac{\text{Block size}}{\text{Record size}}$$

{ When, Space required for block header is negligible, and records are fixed length records }

$$= \frac{(\text{Block size} - \text{Block header size})}{\text{Record size}}$$

{ When block header size is given and records are fixed length records }



Topic : Blocking Factor

* Blocking factor is defined as average number of records per block

$$\text{Blocking (Bf)} = \frac{\text{Block size}}{\text{Avg. Record size}}$$

{ When, Space required for block header is negligible, and records are Variable length records }

$$= \frac{(\text{Block size} - \text{Block header size})}{\text{Avg. Record size}}$$

{ When block header size is given and records are Variable length records }



Topic : IO cost

IO Cost for an access is defined as number of blocks that must be transferred from secondary memory to main memory in order to access that data



Topic : Organization of Records

There are two ways in which records can be organized into the blocks,

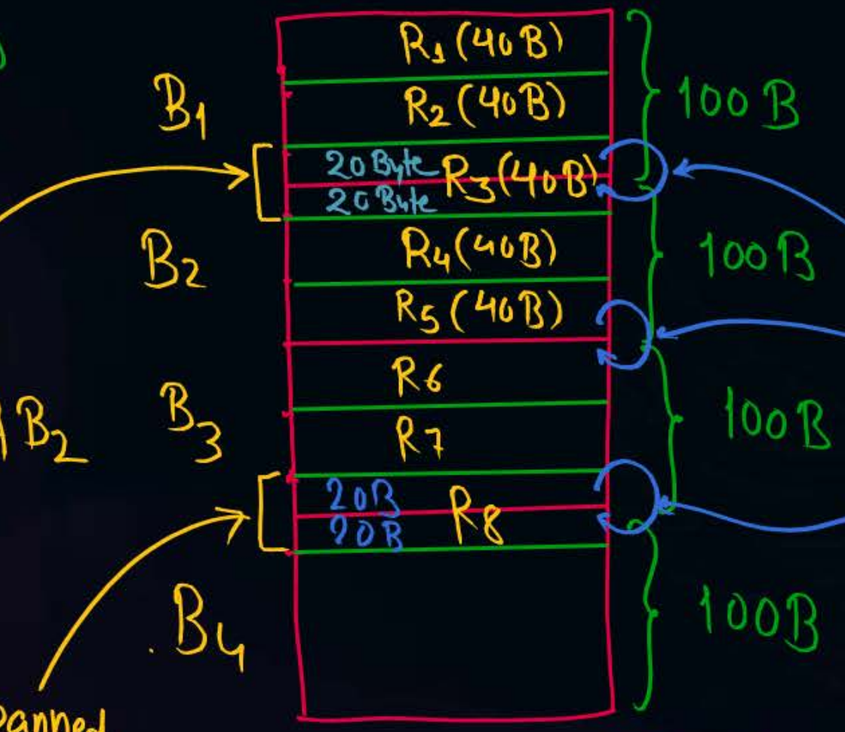
(i) Spanned organization of records

(ii) Un-spanned organization of records.

Topic : Spanned Organization

i.e. A record is allowed span in two blocks.

Let Block size = 100 Byte
& Record size = 40 Bytes



R_3 is Spanned across B_1 & B_2

In order to read the data of spanned records two blocks will be transferred from Sec. mem {∴ More IO Cost}

Spanned

Blocking Factor = 2.5 = $\frac{100}{40}$

{ Space is utilised efficiently, No internal fragmentation }

Block anchor

Block anchor will be used to provide the addr of the next block of file

Topic: UnSpanned Organization

→ In unspanned organization a record must be stored completely within the same block (i.e a record is not allowed to span in two blocks)

let Block size = 100 Byte
& Record size = 40 Bytes



Blocking factor = 2 = $\left\lfloor \frac{100}{40} \right\rfloor$

Disk space is not utilized effectively
(Internal Fragmentation is possible)

{ But in order to read the data of each record only one disk block needs to be transferred (∴ More IO Cost)

Spanned organization

1. No internal fragmentation
2. More IO Cost
3. More suitable with variable length records
4. Blocking factor $= \frac{\text{Block size}}{\text{Record size}}$

Unspanned orgn

1. If is possible.
2. Less IO Cost
3. More suitable with fixed length records.
4. Blocking factor $= \left\lfloor \frac{\text{Block size}}{\text{record size}} \right\rfloor$



Topic : Search-key & Type of File

Search key: Field/Attribute that is being used to search a record in the database file is called search key.

	Eid	- -	- -	- -	Proj-No
B_1	1				7
	5				1
B_2	7				3
	12				9
B_3	13				5
	15				8

Emp

Q: retrieve the record where Eid = x

↓

Search will be Performed based on Eid

i: Eid will be the "Search key"

Q: Retrieve the records of Employee Working on the proj-No = y

↓

Search key will be Proj-No.

Topic : Search-key & Type of File

There are two types of files : (i) ordered file
(ii) Un-ordered file

(i) Ordered-file

If the records of the database file are in sorted order based on search key, then file is called ordered file

* If search key is "Eid" then EMP file is an Ordered file

	Eid	- -	- -	- -	Proj-No
B_1	1				7
	5				1
B_2	7				3
	12				9
B_3	13				5
	15				8

EMP

(ii) Unordered file :-

If the records of the data base file are not in sorted order based on search key, then file is called an unordered file.

* If search key is "Proj-No" then EMP file is an Unordered file



2 mins Summary

- Topic: Database, File and Records
- Topic: Types of Records
- Topic: Blocking Factor
- Topic: Organization of Records
- Topic: Search-key & Type of File
- Topic: Categories of index

Slide

THANK - YOU